V2.4.0 • CODENAME: THE DEVOURER

TECHNICAL PROJECT ABSTRACT & ARCHITECTURE SPECIFICATION

DEVMIND AI — SINGULARITY: EXTREME

Production-Hardened Full-Stack Python & TypeScript AI Repository Intelligence Workstation & Real-Time Relativistic Kerr Black-Hole Simulation

Frontend: React 18, TypeScript 5.6, Three.js / R3F, Custom GLSL, Zustand
Backend: Python 3.13, FastAPI, Pydantic v2, SQLAlchemy 2.x, pgvector / SQLite
Audio: 100% Procedural Web Audio API (6 Real-Time Layers)
AI & Security: Google Gemini SDK, Python AST, Firebase OIDC, Argon2id RBAC

1. EXECUTIVE ABSTRACT

DevMind AI — SINGULARITY: EXTREME (v2.4 — THE DEVOURER) is a production-hardened, full-stack artificial intelligence software engineering workstation built on an asynchronous **Python 3.13 / FastAPI** backend and a real-time **React 18 / TypeScript / Three.js** 3D relativistic Kerr black-hole cosmic simulation. Modern engineering teams struggle to navigate polyglot repositories, audit code for Common Weakness Enumerations (CWEs), defend LLM workflows against adversarial prompt injection, and retain architectural context across sessions. DevMind AI resolves these bottlenecks through ten specialized workstation modules powered by a Python **Hybrid Vector Retrieval-Augmented Generation (RAG)** pipeline, native **Python Abstract Syntax Tree (`ast`)** & multi-language symbol indexing, **SQLAlchemy 2.x** asynchronous persistence (**PostgreSQL + pgvector** / **SQLite**), and **Google Gemini (`google-genai`)** synthesis—all surfaced inside a crystal-glass spatial workspace.

Surrounding the workstation is a deterministic **120 Hz astrophysical simulation and custom GLSL shader engine** inspired by European Southern Observatory (ESO) and NASA Goddard supercomputer visualizations (`AT2019qiz`, `ASASSN-14li`, `SVS 14620`). The visual engine renders a near-extremal rotating Kerr black hole ($a_* \approx 0.94$) featuring a depth-occluding D-shaped Event Horizon shadow, nested relativistic photon rings ($n = 1, 2, 3$), Einstein gravitational starfield lensing, multi-stratum volumetric accretion gas, sequential 6-stage stellar Tidal Disruption Event (TDE) spaghettification, and planetary Roche limit fragmentation—synchronized with a zero-dependency, **6-layer procedural Web Audio API soundscape** and protected by Python-enforced **OAuth 2.0 / OIDC + Argon2id Role-Based Access Control (RBAC)**.

Keywords: Python 3.13, FastAPI, SQLAlchemy 2.x Async ORM, Hybrid Vector RAG, Python Abstract Syntax Tree (`ast`), Google Gemini, TypeScript, Three.js / GLSL, Relativistic Kerr Black Hole, Tidal Disruption Event (TDE), Procedural Web Audio Synthesis, PromptShield, Defense-in-Depth Security.

2. CORE ARCHITECTURAL PILLARS

01 Python AI Repository Intelligence & RAG Plane

- **Singularity Codebase Chat (`gemini_service.py`):** Hybrid cosine-vector + lexical RAG (`vector_engine.py`) with line-level AST symbol citations and multi-turn Google Gemini synthesis.
- **Python AST & Polyglot Indexer (`code_parser.py`):** Multi-file & `.zip` archive ingestion using Python's native `ast` module & regex extractors for symbol, import, and chunk indexing.
- **PromptShield & AI Code Review (`security_engine.py`):** Pre-flight adversarial prompt injection firewall paired with static CWE vulnerability scanning and automated PR review.
- **Orbital Explorer, MemoryMesh & ModelDoctor:** Interactive 3D dependency graph, semantic developer memory graph, live LLM latency/token calibration probes, and AST Markdown doc generator.

02 Relativistic Kerr Black-Hole Visual Engine

- **Kerr Shadow & Photon Rings:** Pure `#000000` D-flattened shadow (`EventHorizon.tsx`) with nested $n = 1, 2, 3$ relativistic photon rings and upper/lower lensed disk arches (`GravitationalLens.tsx`).
- **4-Layer Volumetric Accretion Structure:** Blazing ISCO plasma ring, filamentary MHD-inspired main disk, 3D flared corona strata, and outer cooling dust torus with Doppler beaming & filmic HDR rolloff.
- **Stellar TDE & Planetary Roche Physics:** `120 Hz` fixed-step simulation driving 6-stage stellar spaghettification (44-segment bound fallback ribbon, pericenter shock hotspot, unbound outflow tail) and 3D planetary boulder fragmentation.
- **Gravitational UI & Text Physics:** Trajectory-verified glyph ingestion (`GravityTextEffect`) with native `Ctrl+Z` / `Ctrl+Y` undo reversal and full-viewport `CosmicShockwaveOverlay`.

03 Procedural Web Audio API Cosmic Engine

- **100% Mathematical Synthesis:** Zero external audio files; generates real-time audio across 6 independent layers: *The Void Drone*, *Gravitational Pressure*, *Accretion Storm*, *Stellar Destruction*, *Planetary Destruction*, and *Cosmic Shockwave*.
- **Mastering & Acoustic Safety:** Routed through a `28 Hz` subsonic highpass, `6.5 kHz` lowpass ceiling, `2.2s` dark convolver reverb, and `-4.0 dB` (`10:1`) soft-knee limiter across 5 live presets (including `SINGULARITY` & `CINEMATIC_HORROR`).
- **Autoplay & Visual Sync:** Starts strictly silent until user gesture, mutes on unauthenticated screens, and drives zero-allocation `AnalyserNode` 3D visual modulation.

04 Python Defense-in-Depth Security & RBAC

- **Dual Authentication & Revocation (`auth.py`):** Official Firebase Google OAuth 2.0 / OIDC (`firebase-admin` RS256 ID token verification) + `argon2-cffi` Argon2id password hashing and `token_version` session revocation.
- **Strict RBAC & IDOR Isolation (`main.py`):** Server-enforced `user` / `admin` roles (`require_admin`), `get_owned_repo()` per-user isolation across all endpoints, and request-ID structured audit logging.
- **Upload & Network Hardening (`security_middleware.py`):** ZIP-slip traversal guard, symlink rejection, `50x` compression bomb cap, per-IP/user sliding-window rate limiting (`429 Retry-After`), and strict CSP/HSTS headers.

3. FULL-STACK SUBSYSTEM & MODULE MATRIX

Subsystem Layer	Key Source Modules	Technical Implementation & Capabilities
Python Backend API, RAG & Security Plane	`backend/app/main.py`, `auth.py`, `models.py`, `database.py`, `security_middleware.py`, `services/code_parser.py`, `gemini_service.py`, `vector_engine.py`, `security_engine.py`	**Python 3.13** + **FastAPI** async REST plane; **Pydantic v2** schema validation; fail-closed `validate_startup_config()`; **SQLAlchemy 2.x** async ORM with **PostgreSQL + pgvector** (or local `aiosqlite`); native Python `ast` parser, hybrid vector engine, and server-side `google-genai` key isolation.
3D Relativistic & Celestial Engine	`src/engine/CosmicSimulationManager.ts`, `components/space/EventHorizon.tsx`, `StarCaptureSystem.tsx`, `PlanetCaptureSystem.tsx`	`120 Hz` fixed-timestep integrator with linear render interpolation; zero per-frame heap allocation; 4 stellar spectral classes & 5 planetary archetypes; `PERFORMANCE`, `BALANCED`, and `CINEMATIC` quality tiers.
Procedural Cosmic Audio Subsystem	`src/audio/CosmicAudioEngine.ts`, `CosmicSoundGenerator.ts`, `AudioPerformanceManager.ts`	Single `AudioContext` lifecycle; seeded PRNG noise buffers; exact-once `particleId` & celestial event deduplication (2,048-entry ring); bounded transient polyphony with deterministic node cleanup.
Workstation UI & Glass System	`src/modules/*` (10 Modules), `components/glass/DynamicGlassLighting.tsx`, `GravityTextEffect.tsx`	Crystal-clear dynamic glass (`blur(18px) saturate(165%)`) with shockwave-reactive specular borders; passive bubble-phase deletion tracking preserving native browser undo/redo history.

Python 3.13 + TS	120 Hz	10 Modules	0 CVEs
FULL-STACK CORE	DETERMINISTIC PHYSICS STEP	AI ENGINEERING WORKSTATION	PROD NPM & PIP-AUDIT